ELLA BOURASSA

COMPUTER SCIENCE • MACHINE LEARNING • HUMAN-CENTERED TECHNOLOGY

Brooklyn, NY | (925) 640-8592 | ella_bourassa@yahoo.com
linkedin.com/in/ella-bourassa-144b3130a | ellab130.github.io

PROFILE

M.S. Computer Science student at NYU Tandon with a B.S. in Cognitive Science and a computational emphasis from UC Davis. Interdisciplinary background in machine learning, algorithms, data analysis, and human-centered technology, with experience turning technical work into clear, practical outcomes.

EDUCATION

M.S. Computer Science | New York University, Tandon School of Engineering **Expected May 2028**
Graduate coursework: Design and Analysis of Algorithms I, Programming Languages, Big Data

B.S. Cognitive Science, Computational Emphasis | University of California, Davis **June 2026**
GPA: 3.659 | Selected coursework: Artificial Intelligence, Data Structures, Algorithms, Probability and Statistical Modeling, Linear Algebra, Theory of Computation

TECHNICAL SKILLS

Languages: Python, C, C++, x86 assembly, MATLAB

Libraries: PyTorch, Gymnasium, pandas, scikit-learn

Tools: Git/GitHub, Jupyter, VS Code, CMake, GoogleTest, Valgrind, gdb

Methods: Machine learning, reinforcement learning, statistical modeling, model evaluation, data visualization, data structures and algorithms

SELECTED PROJECTS

ChurnScope - Telecom Churn and Retention Modeling **2026**
Python, pandas, scikit-learn

- Built and evaluated classification models on a telecom churn dataset; the best Random Forest model achieved approximately 0.679 ROC-AUC.
- Converted predicted probabilities into threshold and contact-capacity profit analyses, connecting model performance to customer-outreach decisions.

SARSA Agent for Atari Pong **2025**
Python, PyTorch, Gymnasium ALE Pong-v5

- Implemented a SARSA agent using grayscale 84 x 84 observations, four-frame stacking, and an epsilon-greedy exploration schedule.
- Tracked training reward, compared results with DQN baselines, and documented limitations and potential improvements to representation and training.

EXPERIENCE & LEADERSHIP

Assistant Manager / Shift Lead | Kidplex LLC **Jan-Mar 2025**
San Ramon, CA

- Promoted from front desk staff to shift lead; supervised daily operations, delegated work, and coordinated events of up to 200 guests.
- Built a Google Apps Script inventory tool and analyzed Roller POS sales data to create a dynamic pricing model supporting operational decisions.

Outdoor Gear Rental Associate | UC Davis Outdoor Adventures **Jun 2025-Jun 2026**
Davis, CA

- Guided customers through equipment selection, rentals, and safe gear use while managing payments, reservations, records, and equipment readiness in a fast-paced team environment.

Summer Camp Counselor | UC Davis Youth Programs **May-Aug 2026**
Davis, CA

- Supervised youth participants, led daily activities and group transitions, and communicated with families and coworkers while responding calmly to behavioral, logistical, and safety needs.

Social Media Director | UC Davis Women's Rowing **Apr 2023-Aug 2024**
Davis, CA

- Managed Instagram, Facebook, and TikTok content; created graphics, short-form video, newsletters, and event recaps, and used platform analytics to refine engagement strategy.